Sefik EGRIBOZ (M.Sc., PMP, PSPO, ITIL)

PRINCIPAL SOLUTIONS ARCHITECT

Digital Transformation | Programme Delivery | Enterprise Integration

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PROFESSIONAL SUMMARY

Key results: **200+ integration points across 86 systems** standardized under one governance model at **99.8%** availability | **98%** fewer reported ATC confusion events with AI in production | **EUR 800K** first-year savings

Digital transformation architect and manager with 20+ years in airline IT, driving technology change in operations that cannot stop: legacy modernization off the mainframe (technical driver from day one, as an engineer), an **enterprise integration transformation** onto governed, high-transaction-volume, event-driven, API-first platforms carrying **~50M annual transactions**, and AI in production.

Moved a safety-critical process from heuristics to AI: a fully AI-based callsign safety system for an **NP-hard** Quadratic Assignment Problem across up to ~1,500 daily flights at peak season, alongside a real-time Early Warning System with zero restricted-airspace dispatch incidents since go-live.

Transformation with **delivery accountability**: **PMP-certified**, managed **50+ projects** from small to multimillion-euro operational platform programmes and defined roadmaps for six mission-critical systems (EUR 1M budget); customer-side product owner for Lufthansa Systems NetLine, SchedConnect and flight planning platforms across three carriers, running a multi-vendor RFP (five vendors), POC evaluations and key-user training; rated **exceptional** in the DDI Manager Ready assessment. Treats integration as a change to the operating model, not an interface count.

WORK EXPERIENCE

Career Break and Professional Development | **Istanbul** | **Sep 2024 – Present**

- Planned break after leaving the Turkish Airlines Group, combining family time with structured upskilling: earned **AWS Certified AI Practitioner (AIF-C01)**, **Professional Scrum Product Owner I (PSPO I)** and **AWS Certified Cloud Practitioner (CLF-C02)** in 2026 and continued research on graph-based AI for aviation safety.

Principal Solutions Architect | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2023 – Sep 2024**

Official title: Distinguished Expert, the highest technical grade at Turkish Technology.

- Owned the transformation roadmap and technical direction for the operational platform estate, with transformation governance that tied roadmap, architecture and delivery decisions to **integration dependency management** so new capabilities did not fragment the platform landscape.
- Defined platform roadmaps for **six mission-critical operational systems** serving 1,000 users (~100 concurrent) and 5M annual transactions within a **EUR 1M** budget, backed by critical path analyses reviewed with management.
- Ran the multi-vendor NetLine/Ops replacement selection (**5 vendors, 3 shortlisted, 2 POCs**) and vendor evaluation of flight planning platforms (Jeppesen, CAE formerly FPM, Lido) through RFP/RFI, vendor management and business cases, advising VP-level leadership on build-versus-buy, technology strategy and long-term R&D strategy.
- Protected schedule coherence across two carriers and two reservations systems during AJet's migration from TROYA, the group's reservations and inventory platform, to a separate third-party PSS, running SchedConnect-based schedule synchronization, inter-carrier schedule distribution and schedule change management in a transition architecture.
- Led an **8-person cross-functional team** (project manager, 2 system analysts, 1 business analyst, 4 software engineers) as architecture and design decision owner: defined the implementation approach, mentored engineers hands-on and cleared complex technical blockers, with Scrum Master support as required; rated team members' performance.
- Cut reported ATC confusion events by **98%** (sustained since) on up to ~1,500 daily peak-season flights, replacing the 2011 heuristic season-start assignment with an in-house fully AI-based callsign safety system aligned with EUROCONTROL CSST.
- Reframed callsign assignment as an **NP-hard** Quadratic Assignment Problem and built a scalable machine learning solution on **Graph Neural Networks (GNN)**: **Graph Attention Networks (GAT)** on Neo4j with PyTorch Geometric and feature engineering over phonetic, visual and semantic similarity.
- Established design authority and design reviews bringing architecture decisions under ISO 27001 compliance, software quality management and enterprise architecture governance, and supported regulatory audits and ICAO compliance.
- Shaped integration and technology strategy for next-generation aviation operations platforms as **NetLine Subject Matter Expert** for Turkish Airlines, SunExpress and AJet, and as SME in callsign safety, IOCC and operational control systems.

Senior Integration Architect | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2021 – Jan 2023**

Official title: Senior Expert, Integration Architect.

- Drove the enterprise integration transformation: standardized **200+ integration points across 86 systems** (**~50M annual transactions**) under one integration governance and data governance model, with enterprise messaging and data streaming reference patterns adopted group-wide for interoperability, maintainability, observability and operational reliability.
- Achieved **99.8% availability**, including planned maintenance windows, for mission-critical platforms serving **500+ concurrent users** with **zero unplanned downtime**, by leading cross-functional delivery management through production deployment: coordinating technical dependencies and production readiness across teams, with technical input to steering committee meetings, under high availability, disaster recovery design and SLA/SLO management.
- Extended PSS/DCS integration with the in-house PSS TROYA to irregular operations (IROPS/IRROPS) and disruption management, propagating IOCC decisions to inventory, reservations and departure control as they happened, so disrupted passengers, even those already en route to the aircraft, could re-accommodate online.
- Engineered a real-time Early Warning System across NetLine and FPM: hundreds of automated checks and alarms on planned routes for up to **~1,500 daily** peak-season dispatch legs, and **zero** dispatches into restricted airspace since go-live.
- Modernized the integration landscape onto an event-driven application and data integration platform on Apache Kafka (topic design, schema registry), IBM MQ and REST/SOAP APIs, sustaining **~50M annual transactions** with microservices, enterprise integration patterns, API-first design and lifecycle management, and API versioning and backward compatibility.
- Engineered the estate's reliability semantics: publish/subscribe and asynchronous processing, idempotency in consumers, retries and exponential backoff, timeouts and error recovery, dead-letter handling and poison-message processing, message ordering and durability, and replay of persisted streams for recovery.
- Owned the solution architecture artefacts that kept decisions traceable to a maintained target state architecture: solution blueprints, architecture roadmaps, High-Level and Low-Level Design (HLD/LLD), Architecture Decision Records (ADR) and non-functional requirements (NFR).
- Set the estate's engineering practice: CI/CD architecture and DevOps release automation with Git-based (GitOps) configuration control, with automated testing strategies and performance and security testing gating releases.
- Built a real-time observability tool across all live enterprise integrations, using structured logging, correlation IDs and distributed tracing; directed delivery of Grafana dashboards, metrics and alerting to support proactive issue detection.

Project Manager, Flight Operations | **Turkish Airlines** | **Istanbul, Turkiye** | **Jun 2014 – Jan 2021**

Functional title: Lead Solutions Architect and Platform Owner.

- Owned delivery scope, implementation planning, delivery dependencies, vendor activities, user acceptance testing (UAT), risk tracking and operational handover for multi-system releases across scheduling, flight planning, IOCC and downstream enterprise systems, through to production.
- Drove technical direction, integration and platform modernization for **multimillion-euro** airline operational systems (scheduling, slot management, flight planning, IOCC) as lead architect with delivery accountability: system and business analysis, software architecture and hands-on coding through release, operational handover and 24/7 production under ITIL.
- Designed and built **TKAODS** (Airline Operational DataStore) and evolved **OFPDS** (Operational Flight Plan DataStore, live since 2010), **Tier-0** integration layers in front of the Lufthansa Systems NetLine suite (NetLine/Sched, NetLine/Slot, NetLine/OPS & Base) and flight planning that became the **single source of truth** for schedule and flight data.
- Owned enterprise-wide flight plan and flight data distribution for planning, operational and post-flight periods as the single integration point for **eleven main downstream domains**: scheduling, crew, load control, PSS/DCS interfaces, cargo, maintenance, DWH, revenue and cost accounting, SAP finance, HR/payroll and regulatory reporting.
- Processed **~100K update records a day** (peak-season average) for up to **~1,500 flights a day**, 24/7 and near-real-time, through source-to-target mappings, ETL/ELT pipelines and integration rules that consolidated, transformed and enriched every update across summer and winter seasons under data quality and metadata controls.
- Designed batch, near-real-time and real-time distribution patterns that absorbed **~22.5M** flight-data changes a year (~450K annual flights, ~50 changes per flight) before fan-out to large-scale distributed data processing and analytics, with consumer-specific horizons from a **rolling three-day window** to a **90-day window**.
- Fed operational changes (delays, cancellations) to passenger-notification systems and flight data to loyalty and disruption hotel systems, so passenger notifications, loyalty and disruption hotel arrangements ran on the same data as operations.
- Designed the enterprise operational data and integration architecture on the AODB (Airline Operational DataStore) pattern, standardizing flight, leg, schedule and operational flight-plan data airline-wide: logical data models, entities and relationships, canonical data structures, and the ESB pattern applied twice.
- Evolved both Tier-0 platforms for **ten years** (2014 – Sep 2024) as de facto product owner with final authority over feature prioritization and architectural evolution, running agile delivery in **Jira** and **Confluence**: user stories with acceptance criteria, backlog management, sprint planning and Definition of Ready/Done.
- Coordinated operational stakeholders and cross-squad dependencies for two Tier-0 platforms, ran critical path analyses reviewed with management and gave technical input in steering committee meetings, carrying releases through production under ITIL-aligned **Site Reliability Engineering (SRE)** practices and continuous monitoring.

- Managed NetLine's master and reference data (NetLine/Base) with the team: built and maintained the tables where users edit it and trained key users in master data management (MDM) to support user adoption and organisational change, keeping master data owned at its source.
- Built a business intelligence reporting layer into both platforms on the same enriched data and fed it back into NetLine and the flight planning system, so enterprise reporting and operations shared one version of the data.

Integration Solution Design Lead | **Turkish Airlines** | **Istanbul, Turkiye** | **Jan 2007 – Jun 2014**

Official title: Acting Team Leader.

- Led a **6-person team** through the lifecycle of **100+ integrations** (requirements and system analysis, design, development, testing, documentation), harmonizing project plans and tracking cross-functional milestones per project. As Acting Team Leader, rated team members' performance and interviewed technical candidates, with hiring decisions made by others.
- Delivered **EUR 800K** in first-year savings with a multi-currency overflight cost engine, launched in 2012 and built on a data pipeline of country entry/exit timestamps and segment distances.
- Earned PMP and ran scope definition, executive (C-level) stakeholder management, customer relationship management, vendor management, risk tracking, implementation planning and operational handover.
- Headed solution design and delivery for flight operations, crew and flight-planning integrations across global aviation vendors, building multithreaded real-time integration frameworks (2010) and contributing to an in-house Enterprise Service Bus (ESB) on Service-Oriented Architecture (SOA).
- Introduced SchedConnect in 2007 for inter-carrier schedule enrichment and distribution, establishing recurring schedule data exchange with partner carriers under IATA data standards.
- Architected the schedule-to-PSS (TROYA) message integration under IATA SSIM Chapter 4 (SSM) and Chapter 5 (ASM) over Type B messaging, keeping inventory and reservations in step with cancellations, re-timings, equipment and routing changes, reinstated flights and disruption reason codes.
- Developed the first heuristic Callsign Similarity & Assignment System (phonetic, visual and semantic similarity), delivered in 2011: the groundwork later rebuilt as the graph-based AI solution once reclassified as NP-hard.
- Commended in writing in 2010 by Turkish Airlines' **Chief Pilot and Head of Flight Operations** for flight operations projects, the callsign project first among them, and personally by the **General Manager and CEO**.

Software and Integration Engineer | **Turkish Airlines** | **Istanbul, Turkiye** | **Apr 2000 – Jan 2007**

Official title: Analyst Programmer.

- Drove the technical work of the first major legacy modernization from day one as an implementation team member, moving schedule and operations control off the IBM mainframe (ADABAS/Natural) onto Oracle on Unix (AIX) and NetLine.
- Contributed to the first production implementation of the Lufthansa Systems NetLine suite (NetLine/Sched, NetLine/Slot, NetLine/OPS & Base), the foundation of fleet growth from **60 to 500+ aircraft** and passengers from 10M (2000) to 85M (2024).
- Cut costs by **EUR 100K** a year by designing and building the timetable publication system from scratch: worldwide airline schedules and partner codeshare flights in, a downloadable timetable on the corporate website out (Java, C, PL/SQL).

CORE COMPETENCIES & TECHNICAL SKILLS (hands-on)

Transformation and Programme Leadership: Digital and technology transformation, transformation governance, programme management, delivery governance, legacy modernization, roadmap definition, executive (C-level) stakeholder management, customer consulting and workshop facilitation.

Change, Adoption and Benefits: Stakeholder alignment, user readiness through UAT and key-user training, operational handover, multi-carrier platform adoption; benefits realization through delivered and measured outcomes (EUR 800K first-year savings, 98% fewer reported ATC confusion events, 99.8% availability).

Technical Platforms, Data & Architecture: Enterprise Data Architecture, Data Modeling, Canonical Data Model, Operational Data Store (ODS), Data Integration Architecture, Data Pipeline Architecture, Event-Driven Architecture, API Architecture, Apache Kafka, IBM MQ, Enterprise Service Bus (ESB), REST and SOAP APIs.

Aviation Operations and Airline IT: Integrated Operations Control Center (IOCC), operational and tactical flight planning (Jeppesen; FPM, now CAE Flight Planning Solutions; Lido), schedule management; DWH, SAP and MRP interfaces.

IATA Schedule Data Standards: SSIM Chapter 7 (Presentation and Transfer of a Schedule Data Set); schedule-to-PSS distribution across three group carriers.

Analytics & Graph / Advanced Analytics and AI: Neo4j and graph databases, PyTorch and PyTorch Geometric, MLOps, data quality and metadata management.

Engineering Platforms & Languages: Oracle DB, Docker and containerization, Linux (RHEL) and Unix administration (AIX, HP-UX), application administration on Windows Server; Languages: Python, Java, C, PL/SQL, shell scripting (Bash, KSH), YAML and JSON; Git and Jenkins; UML; multithreaded real-time and database-driven systems; incident analysis and performance tuning.

EDUCATION

• Ph.D. (coursework)	Business Administration (Finance)	Yeditepe University	Istanbul
• M.Sc.	Management Engineering; thesis on data warehousing for mass customization	Istanbul Technical University	Istanbul
• B.Sc.	Computer Science Engineering	Istanbul University	Istanbul

LANGUAGES

• English	C1	Full Professional
• German	A2	Studying towards B1
• Turkish		Native

CERTIFICATIONS

• AWS Certified Cloud Practitioner (CLF-C02)	Amazon Web Services (AWS)	2026
• AWS Certified AI Practitioner (AIF-C01)	Amazon Web Services (AWS)	2026
• Professional Scrum Product Owner I (PSPO I), Credential ID 1347372	Scrum.org	2026
• AI and Cloud Foundations: OCI AI Foundations; Agentic AI Foundations; IBM SkillsBuild; PMI AI (CPMAI, GenAI)	Oracle, IBM, PMI	2025
• Docker Foundations Professional Certificate	Docker	2025
• Manager Ready (DDI) leadership simulation: exceptional rating	TAG Taurus Group	2024
• ITIL v3 Foundation, Candidate No. 990453483	BCS, The Chartered Institute for IT	
• Project Management Professional (PMP), Credential ID 1225730	PMI	
• Lufthansa Systems SchedConnect vendor training (two weeks)	Lufthansa Systems	